1. What is the main goal of approximation algorithms?

   A To find the optimal solution
   B To find a solution close to optimal
   C To reduce the running time
   D To increase the memory usage

2. Which algorithm is used for vertex cover with an approximation guarantee of 2?

   A Greedy algorithm
   B Select an arbitrary edge and add both endpoints to the solution
   C Dynamic programming
   D Brute force

3. What is the approximation guarantee of the greedy algorithm for load balancing?

   A 2
   B 3
   C 4
   D 5

4. What is the goal of the load balancing problem?

   A To minimize the average load
   B To minimize the maximum load
   C To maximize the minimum load
   D To maximize the average load

5. Which algorithm is used for load balancing?

   A Greedy algorithm
   B Dynamic programming
   C Brute force
   D Greedy algorithm assigning jobs to machines with the smallest load

6. What is the improved approximation guarantee for the load balancing algorithm?

   A 2
   B $(2 - \frac{1}{m})$
   C 3
   D 4

7. What is the definition of an α-approximation algorithm?

   A A polynomial-time algorithm that returns a solution within a factor α of optimal
   B A polynomial-time algorithm that returns an optimal solution
   C A polynomial-time algorithm that returns a solution within a factor $\frac{1}{\alpha}$ of optimal

D A polynomial-time algorithm that returns a solution within a factor α^2 of optimal

8. What is the name of the problem where we want to return a subset of vertices that covers all edges in a graph?

   A Vertex cover

   B Load balancing

   C Scheduling

   D Approximation algorithms

9. What is the goal of the vertex cover problem?

   A To find a subset of vertices that covers all edges

   B To find a subset of edges that covers all vertices

   C To find a subset of vertices that covers some edges

   D To find a subset of edges that covers some vertices

10. What is the name of the problem where we want to assign jobs to machines to minimize the maximum load?

    A Load balancing

    B Vertex cover

    C Scheduling

    D Approximation algorithms